Conformal Blocks and Crossing Symmetry

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Abstract

Here is a note on a AMAZING topic of 2 dimensional CFT developed in the 80s.

Contents

1 Introduction

1.1 Overview

Since the BPZ paper gives a concrete definition of the conformal blocks, we want to control the behavior of these special functions. A series of papers by Moore and Seiberg in the late 80s and early 90s give an answer. Without concretely calculating the conformal blocks, they show that these function have certain crossing properties, controled by series of duality matrices. More amazingly, these duality matrices can be explicitly calculated. This gives a explicit way of bootstrapping the 2D CFT structure constants.

Moreover, apart from the field theory application. The crossing properties of conformal blocks forms a beautiful mathematical structure, so called the modular tensor category. This gives us a deep connection between 2D CFT and 3D TQFT. In fact, it can be shown explicitly from the Chern-Simons / WZW correspondence, which is a prototype of holographic duality.

More amazingly, the conformal blocks can also be view as a holomorphic section of a vector bundle over the moduli space of Riemann surfaces. This gives a deep connection between 2D CFT and complex geometry. This point of view also gives us insights on the quantization of Teichmuller space, which is a key step in understanding AdS3 gravity.

In this note, I'll mainly focus points:

1. duality matrices and crossing of conformal blocks.
2. Underlining Modular Funtor and Modular Tensor Category Mathematical Structure.

1.2 Axiomatic CFT

We first review the BPZ axioms of 2D CFT.

2 CVO Construction of Conformal Blocks

To understand the crossing properties of virasoro or more general Kac-Moody (KM) conformal blocks, we need tools for calculating or at least representing these special functions. The Chiral Vertex Operator (CVO) construction is a powerful method for doing this.

Modernly, mathematicians may prefer some more abstract construction and CVO are not quite commonly mentioned. However, for physicist, I think its an important tool for understanding the structure of conformal blocks and their crossing properties.

2.1 BPZ Conformal Blocks

Let's first review the basic definition of BPZ conformal blocks. Mathematicians have generalized this definition to more general Riemann surfaces, but as far as we concern, we only need definitions on punctured spheres and torus.

2.1.1 4-Punctured Sphere Blocks

2.1.2 Torus Blocks

2.2 Chiral Vertex Operators

2.3 Mathematical Reformulation

3 Duality Matrices

Bibliography